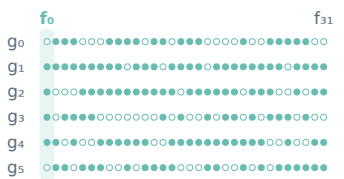
A Attribute-defined object prediction

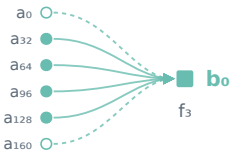
1 Assign categorical attributes



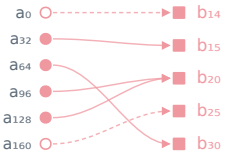
2 Create two matched relations

Attribute rule

same f_0, six different g



Balanced random



3 Repeat & shuffle sentences

Attributes · all 224 entities ×128

x a_0 f_0 g_0 x

x b_0 f_3 g_0 x

Attribute rule · 128 triples ×16

x a_{32} r_0 b_0 x

Random control · 128 triples ×16

x a_{32} r_1 b_{15} x

B Attribute equality & symmetric lookup

64 entities → binary pair labels

1 Assign each entity 10 bits

Attribute

	0	1	2	3	4	5	6	7	8	9
s_0	1	1	0	0	1	1	0	1	1	1
s_1	1	0	0	1	1	1	0	1	0	1
s_5	1	0	0	0	1	0	1	0	1	0
s_2	0	1	1	0	1	0	1	0	0	0
s_3	0	1	0	0	1	1	1	1	1	0
s_4	0	1	0	0	0	0	1	0	0	1

⋮

6 of 64 entities shown

2 Label every entity pair

10 equality relations

s_0 1 = 1 s_1

same bit → y

	s_0	s_1	s_5	s_2	s_3	s_4
s_0	y	y	y	n	n	n
s_1	y	y	y	n	n	n
s_5	y	y	y	n	n	n
s_2	n	n	n	y	y	y
s_3	n	n	n	y	y	y
s_4	n	n	n	y	y	y

s_0, s_1 → y

r_0 shown

1 random relation

link → y

no link → n

	s_0	s_1	s_5	s_2	s_3	s_4
s_0	y	n	y	y	y	y
s_1	n	y	n	n	n	y
s_5	y	n	y	y	y	y
s_2	y	n	y	y	y	n
s_3	y	n	y	y	y	y
s_4	y	y	y	n	y	y

s_0, s_1 → n

31 links + self per entity

3 Write training sentences

Attributes · forward + inverse

s_0 $attr_0$ → 1

$attr_0$ 1 → s_0

Same attribute

r_0 s_0 s_1 → y

Random lookup

r_{10} s_0 s_1 → n

Joint training after the equality warm-up

10 equality relations 1 random relation

Random: 10× each equality relation